

Due on Wednesday, May 6th.

If pictures are needed/relevant, please provide them with your solutions.

Marking: either only problem 1, or problems 2-4, will be marked.

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1. An electromagnetic system in vacuum (no charges or currents are present!) is described by the fields

$$\begin{aligned} \mathbf{E} &= E_0 \exp \left[-\frac{(x - wt)^2}{2L^2} \right] \hat{j} \\ \mathbf{B} &= (E_0/w) \exp \left[-\frac{(x - wt)^2}{2L^2} \right] \hat{k} \end{aligned} \quad \left| \begin{array}{l} L \text{ and } w \text{ are positive constants.} \\ \exp[u] \text{ is common notation for} \\ \text{the exponential function } e^u. \end{array} \right.$$

- (a) **[8 pts]** Show that these fields obey Maxwell's third equation (which concerns the curl of the electric field).
- (b) **[10 pts]** The fields should also obey Maxwell's fourth equation in vacuum. Find the value of w for which this works.
- (c) **[not marked]** Look up the properties of the gaussian function (e.g., wikipedia). You should be able to tell the center and the width of a gaussian by looking at its form.
- (d) **[6 pts]** Sketch plots of the magnitude of the electric field as a function of the position, at the time instants $t = 0$, $t = 2L/w$, and $t = 4L/w$. (You can think of these as 'snapshots' at different points of time.)
Please submit a hand-drawn sketch, not a computer printout.
In which direction is our electromagnetic pulse traveling?
- (e) **[6 pts]** Calculate the Poynting vector $\mathbf{S}$. In case it has not been treated in class yet: $\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}$.
- (f) **[4 pts]** The Poynting vector gives the 'energy current density' for the flow of electromagnetic energy. Explain whether the calculated direction of $\mathbf{S}$ makes sense.
- (g) **[8 pts]** Calculate the energy density for electromagnetic fields,

$$u = \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right)$$

for the given fields.

- (h) [8 pts] Conservation of electromagnetic energy in vacuum is encoded in the relation (analogous to the continuity equation for charge)

$$\nabla \cdot \mathbf{S} + \frac{\partial u}{\partial t} = 0.$$

Find out whether this is satisfied for our fields.

2. [12 pts] Using Maxwell's equations in vacuum and the definitions of $\mathbf{S}$ and u , derive the relation $\nabla \cdot \mathbf{S} = -\partial u / \partial t$. You may need the vector identity

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$$

3. Consider an electromagnetic wave travelling through empty space described by the electric and magnetic fields

$$\mathbf{E} = 3\alpha \cos\left(\frac{1}{L}(y - ct)\right) \hat{i}, \quad \mathbf{B} = \mathbf{G} \cos\left(\frac{1}{L}(y - ct)\right)$$

where α and L are positive constants and $\mathbf{G}$ is a constant vector.

- (a) [2 pts] In which direction is this wave traveling?
- (b) [12 pts] Find the magnitude (in terms of α) and the direction of the constant vector $\mathbf{G}$. You might need to use one of Maxwell's equations.
- (c) [4 pts] What is the wavelength and the frequency of this wave?
4. An electromagnetic wave has an electric field given by

$$\mathbf{E} = E_0 \cos\left(\frac{2\pi c}{\lambda}t\right) \sin\left(\frac{2\pi z}{\lambda}\right) \hat{i}$$

where λ is a positive constant.

- (a) [10 pts] Use Maxwell's third equation (Faraday's law in differential form) to calculate the associated magnetic field $\mathbf{B}$. You can assume the time-independent additive term (constant of integration) to be zero.
- (b) [2 pts] Is this wave traveling? In which direction?
- (c) [4+4 pts] Calculate the Poynting vector and explain the direction of energy flow using your result.